Predicting Personal Loan Approvals

A Machine Learning Approach to Credit Risk

Background & Objectives

The Transition to Data-Driven Decisions

Traditional loan approval processes involve manual assessments of credit history and income stability. While effective, they are inherently time-consuming, subjective, and prone to inconsistencies.

With the rapid rise of digital banking, machine learning models can now analyze complex, non-linear relationships across vast amounts of customer behavioral data.

The Impact: Accurate ML models enable faster, more consistent decisions—reducing default risk while optimizing the customer experience through quicker approvals.

The Core Objective

"Develop a machine learning model that can accurately predict whether a customer is likely to be approved for a personal loan based on their demographic, financial, and behavioral attributes with primary focus on the precision."

Experiment Data Overview

307,511

ANONYMISED CLIENT RECORDS

122

DISTINCT DATA FEATURES

"Home Credit Default Risk"

Sourced from Kaggle, this dataset mirrors real-world, high-dimensional data utilized by modern financial institutions to infer unobservable credit risk.

Data composition includes:

- **Socio-Demographic:** Gender, Education, Family Status
- **Financial Indicators:** Total Income, Credit Amount, Annuity
- **Behavioral Data:** External Bureau Scores, Days Employed

Feature Engineering Strategy

Financial Ratios

Generated new compound features like Debt-to-Income and Loan-to-Income ratios. These ratios are superior indicators of a client's actual repayment capacity compared to absolute values.

Temporal Data

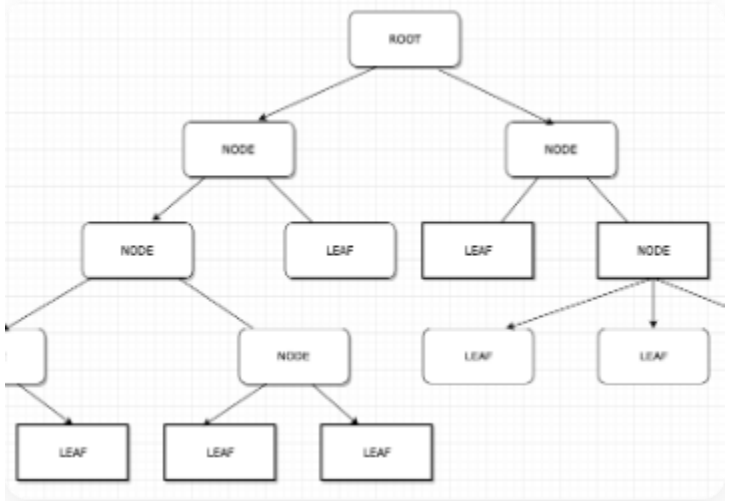
Converted raw temporal columns (e.g., DAYS_BIRTH, DAYS_EMPLOYED) into highly interpretable human-centric metrics like Client Age and Years of Employment to capture stability.

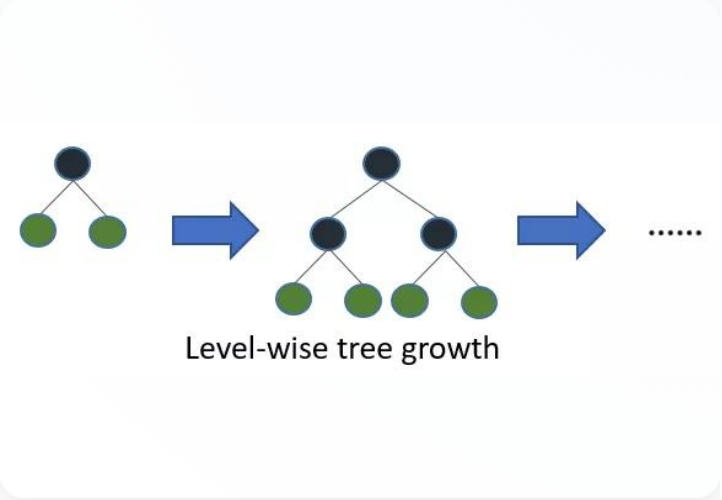
External Sources

Leveraged and aggregated external bureau scores (EXT_SOURCE 1, 2, 3), which are empirically proven in academic literature to be among the strongest standalone predictors of default.

The Ensemble Model Architecture

A Stacking Classifier finalizing predictions with Logistic Regression.





Random Forest

A robust base estimator excelling at handling complex, non-linear relationships across the 122 features without overfitting.

XGBoost

An extreme gradient boosting algorithm optimized for execution speed and high precision execution on structured data.

LightGBM

A highly efficient tree-based learning algorithm providing low memory usage and high-speed processing on large datasets.

Model Tuning Approaches

The initial model achieved 94.08% accuracy. However, since 92% of historical applicants had no financial difficulties, a baseline "approve all" model naturally sits at 92%. Thus, we implemented targeted tuning to maximize **Precision**.

Hyperparameter Tuning on Base Trees Iterated across parameters (e.g., number of estimators, leaf nodes) ranging from 5 to 95. Peak model performance was stabilized when key parameters were set around a value of 45.

Balancing Positive and Negative Cases Addressed data imbalance by down-sampling to an even 498 positive and 498 negative cases. Testing revealed this approach significantly degraded performance (Accuracy dropped to 66%), leading to its rejection.

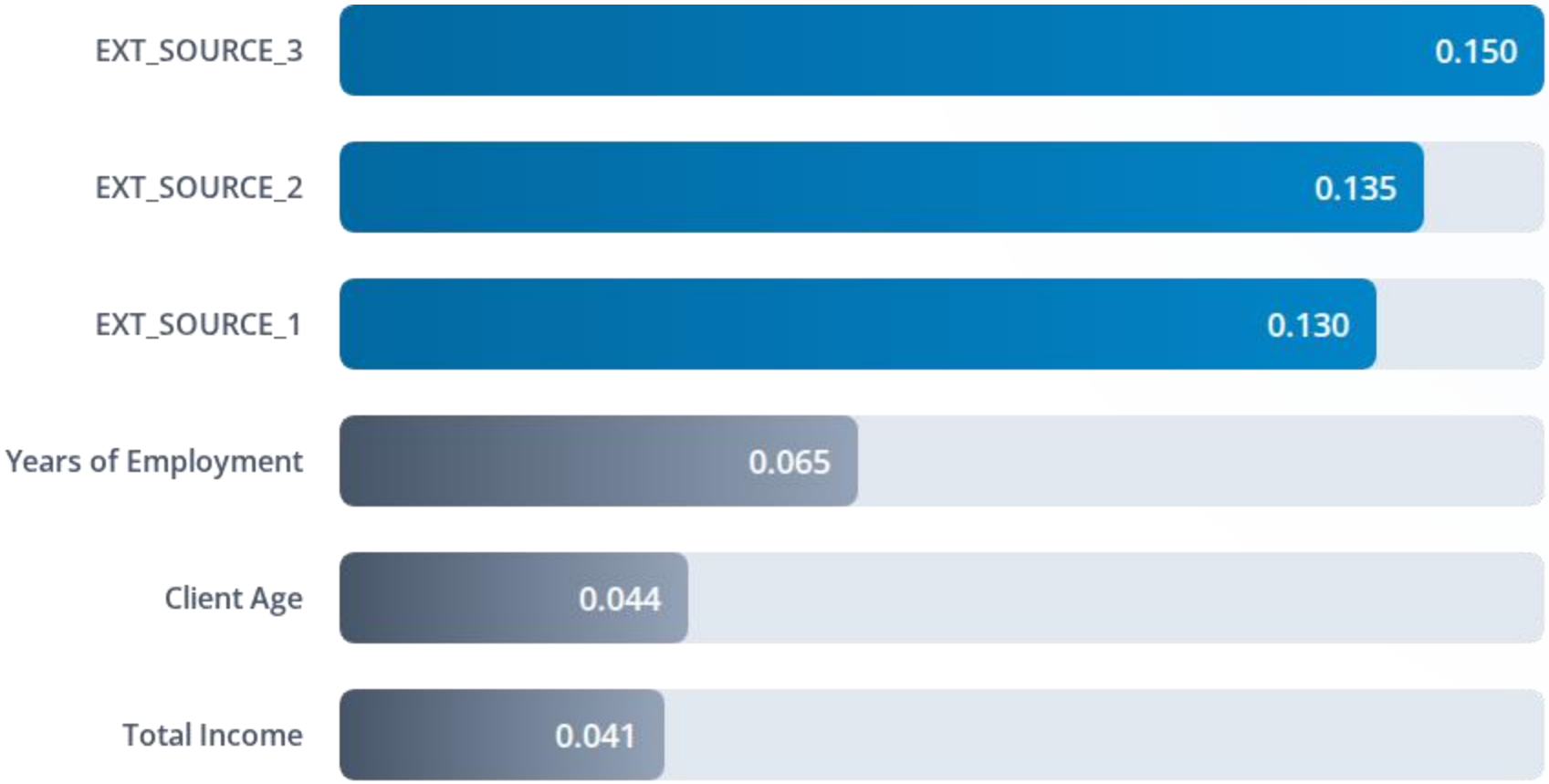
Decision Boundary Threshold Tuning Recognized that financial risk decisions are rarely 50/50. Evaluated predictive probability thresholds from 0.1 to 0.9 to find the optimal cut-off for loan approvals.

Threshold Tuning Results

Threshold	Training Acc.	Training Prec.	Validation Acc.	Validation Prec.	Testing Acc.	Testing Prec.
0.1	95.54%	95.42%	94.86%	94.86%	94.16%	94.16%
0.3	100.00%	100.00%	94.86%	94.86%	94.16%	94.16%
0.5 (Optimal)	100.00%	100.00%	**94.95%**	**94.94%**	**94.16%**	**94.16%**
0.8	100.00%	100.00%	94.51%	95.08%	93.82%	94.45%
0.9	99.96%	100.00%	89.64%	95.93%	**88.33%**	94.97%

Analysis: Model performance hits an optimal balance between 0.4 and 0.5. Raising the threshold past 0.6 marginally increases precision but causes a severe degradation in overall testing accuracy.

Top Correlated Predictive Features



Key Observation: While no single feature strictly dominates the model, the three External Credit Bureau scores are by far the strongest predictors of loan default risk, completely validating established academic consensus (Ranjan Lenka et al., 2024).

Conclusion & Future Outlook

Model Efficacy

Tree-based stacking algorithms proved exceptionally well-suited for this binary classification problem, yielding an operational ~94% precision rate.

Academic Synergy

Insights drawn from existing academic research were invaluable, substantially narrowing down the scope of feature engineering.

Future Enhancements

The model's business value can be further amplified by integrating the predicted default probability with the *requested loan amount*. A high-value loan carries significantly greater financial exposure than a smaller loan, even at identical default probabilities.

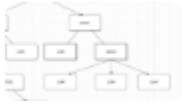
Questions & Discussion

Thank you for your attention.

Image Sources

https://www.mckinsey.com/~/media/mckinsey/industries/financial%20services/our%20insights/financial%20data%20and%20markets%20infrastructure%20positioning%20for%20the%20future/thumb-gettyimages-2177547021.jpg

Source: www.mckinsey.com



https://i.sstatic.net/NUqpT.png

Source: stats.stackexchange.com

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Source: stockcake.com

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Source: 5gstore.com

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Source: www.vecteezy.com